C1 Houlsby bottleneck summarizer: Methodology and Evidence

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Status. Post-run methodology reconstruction grounded in archived source and reports. This document is not an executed historical notebook.

Research question

Reuse the Phase 1 C1 legal encoder while changing a small bottleneck module for summarization. This tests a different Phase 1 lineage and adaptation family against B1 fusion under the same summarization contract.

Architecture and mechanism

Houlsby-style adapters insert down-project, nonlinear transform, up-project residual modules in the encoder. Recurrent readout tuning and the generation head map case evidence into the summary task.

Measured result

Scored submission	1030778, 1030779, 1030987
Codabench BERTScore F1	0.7700
Archived local metric	-1.9414 validation negative NLL; BERTScore not measured in this report

The Codabench BERTScore F1 was 0.7700. The run report selects epoch 3 using negative summary NLL; that is a training selection metric, not the hidden score.

Limitations and next check

Different backbones and adaptation methods change together, so a single score cannot isolate the benefit of Houlsby adapters. Local, pilot, and hidden metrics are deliberately kept distinct. A single scored run does not establish a causal architecture effect or statistical significance.

Source and reproducibility

Original source modules: c1_houlsby.py, phase6_data.py, train_phase6.py. SHA-256 hashes, the original report, and the prediction ZIP receipt are in the companion provenance file. The explanatory notebook was written after the historical run; original execution output is never implied. Reproduction requires authorized course data, the Phase 1 parent weights/tokenizer, and the dependencies in the archived source.

Documentation disclosure

AI assistance was used to organize the archived evidence and draft this explanatory document. All numerical claims above come from saved run records or the provided finished-submission screenshot; missing evidence is labelled as such.